

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 9

Question:

1. Evaluate

$$\oint_C (3y \, dx + 2x \, dy)$$

where C is the boundary of $0 \leq x \leq \pi, 0 \leq y \leq \sin x$. (1)

2. Find the flux of the field $\mathbf{F}(x, y, z) = z^2\mathbf{i} + x\mathbf{j} - 3z\mathbf{k}$ outward through the surface cut from the parabolic cylinder $z = 4 - y^2$ by the planes $x = 0, x = 1$, and $z = 0$. (2)

Please note: There is no fractional marking. Only integer marking.